EXP NO: 3 Experiments on CISCO PACKET TRACKER (Simulation tool)

AIM: To study the packet tracker tool installation and User interface Overview.

c) To understand environment of CISCO PACKET TRACKER to design simple network.

INTRODUCTION:

A simulator, as the name suggests, simulates network devices and its environment. Packet Tracer is an exciting network design, simulation and modeling tool.

1. it allows you to model complex systems without the need for dedicated equipment.

2. it helps you to practice your network configuration and trouble shooting skills via computer or -on an ios based mobile device.

3. it is available for both the linux and windows desktop environments.

4. Protocols in Packet Tracer are coded to work and behave in same way as they would on real hardware.

INSTALLING PACKET TRACER:

To download packet Tracer, go to https://www.netacad.com and login with your cisco networking, then, click on packet tracer graphic and download the package for your operating system.

WINDOWS:

Installation in windows is pretty simply and straight forward; the setup comes in a single file named packet tracer - setup 6.01 ex, open the file to begin the setup wizard, accept the license agreement, choose a location and start installation.

Linux:

Linux users with an ubuntu / Debian distribution should download the file for ubuntu, and those using fedora / Redhat / centros must download the file for fedora. Grant executable permission to this file by using chmod, and execute to begin the installation; packet-Tracer 601-1386-installer-rpm-bin.

./ packet Tracer 601-1386-installer-rpm.bin

USER INTERFACE OVERVIEW:

The layout of the packet Tracer is divided into several components. The interface are as follows, match the numbering with explanations.

1. menu-bar - This is common menu found in all software applications; it is used to open, save; print, change preferences and so on.
2. main toolbar - This bar provides shortcut icons to menu options that are commonly used such as open, save, zoom, undo and redo and on the right hand side is an icon for entering network information for the current networks.
3. Logical / physical workspace tabs - These tab allow you to toggle between logical and physical work areas.
4. Workspace - This is the area where the topologies are created and simulations are displayed.

5. common tool bar - This toolbar provides control for manipulating topologies, such as selects move layout, place note, delete, inspect, resize shape and simple PDU.

6. Real-time / simulation tabs - These tabs are used to toggle between the real and also provided to control the time and to capture the packet.

7. Network component box - this component contains all of network and end devices available with packet tracer, further divided into two areas, Area 7a: Selection box - These area contains device categories. Area 7b: device-specific selection box displays.

8. User-controlled packet - box can create highly customized packets from this area, and results are displayed as a list.

d) Analyse the behaviour of network devices using cisco packet Track simulator:

1. From the network component box, click and drag highly components:

   a) 4 Generic PCs and one HUB

   b) 4 Generic PCs and one switch.

2. click on Connections:

   a) click on copper straight - Through cable.

   b) Select one of the PC and connect it to HUB using cable.

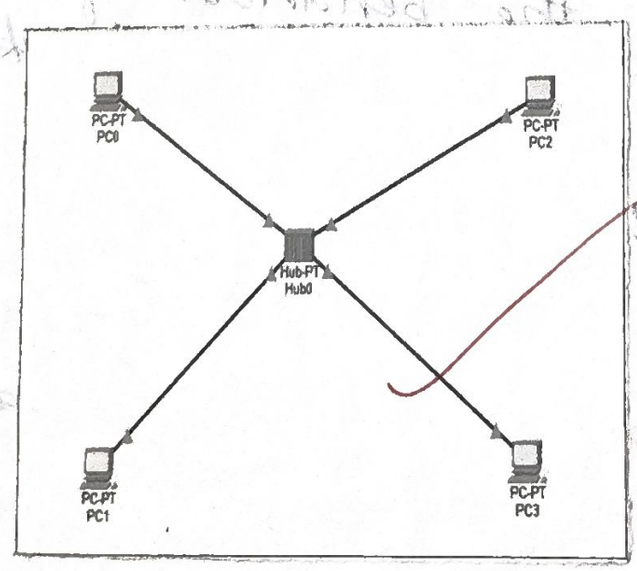
c) similarly connect 4 PCs to the switch using copper straight-through cable.

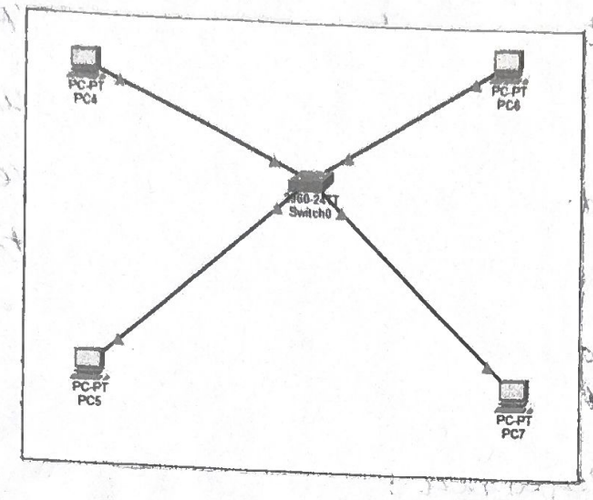
3) click on the PCs connected to hub, go to desktop tab, click on IP configuration, and enter an IP address and subnet mask. The default gateway and not needed there are only two end devices in the network.

4. Observe the flow of PDU from source Pc to destination Pc by Realtime mode of simulation

5. Repeat step #3 to step #5, connected to the switch.

6. observe how HUB and switch are forwarding the PDU and write your observation and conclusion about behaviours of switch and HUB.





Result:

The study of cisco packet tool installation is successfully verified.